A Systematic Review and Meta-analysis of Childhood Leukemia and Parental Occupational Pesticide Exposure

Objectives We conducted a systematic review and meta-analysis of childhood leukemia and parental occupational pesticide exposure. Data sources Searches of MEDLINE (1950–2009) and other electronic databases yielded 31 included studies. Data extraction Two authors independently abstracted data and assessed the quality of each study. Data synthesis Random effects models were used to obtain summary odds ratios (ORs) and 95% confidence intervals (CIs). There was no overall association between childhood leukemia and any paternal occupational pesticide exposure (OR = 1.09; 95% CI, 0.88–1.34); there were slightly elevated risks in subgroups of studies with low total-quality scores (OR = 1.39; 95% CI, 0.99–1.95), ill-defined exposure time windows (OR = 1.36; 95% CI, 1.00–1.85), and exposure information collected after offspring leukemia diagnosis (OR = 1.34; 95% CI, 1.05–1.70). Childhood leukemia was associated with prenatal maternal occupational pesticide exposure (OR = 2.09; 95% CI, 1.51–2.88); this association was slightly stronger for studies with high exposure-measurement-quality scores (OR = 2.45; 95% CI, 1.68–3.58), higher confounder control scores (OR = 2.38; 95% CI, 1.56–3.62), and farm-related exposures (OR = 2.44; 95% CI, 1.53–3.89). Childhood leukemia risk was also elevated for prenatal maternal occupational exposure to insecticides (OR = 2.72; 95% CI, 1.47–5.04) and herbicides (OR = 3.62; 95% CI, 1.28–10.3). Conclusions Childhood leukemia was associated with prenatal maternal occupational pesticide exposure in analyses of all studies combined and in several subgroups. Associations with paternal occupational pesticide exposure were weaker and less consistent. Research needs include improved pesticide exposure indices, continued follow-up of existing cohorts, genetic susceptibility assessment, and basic research on childhood leukemia initiation and progression.